

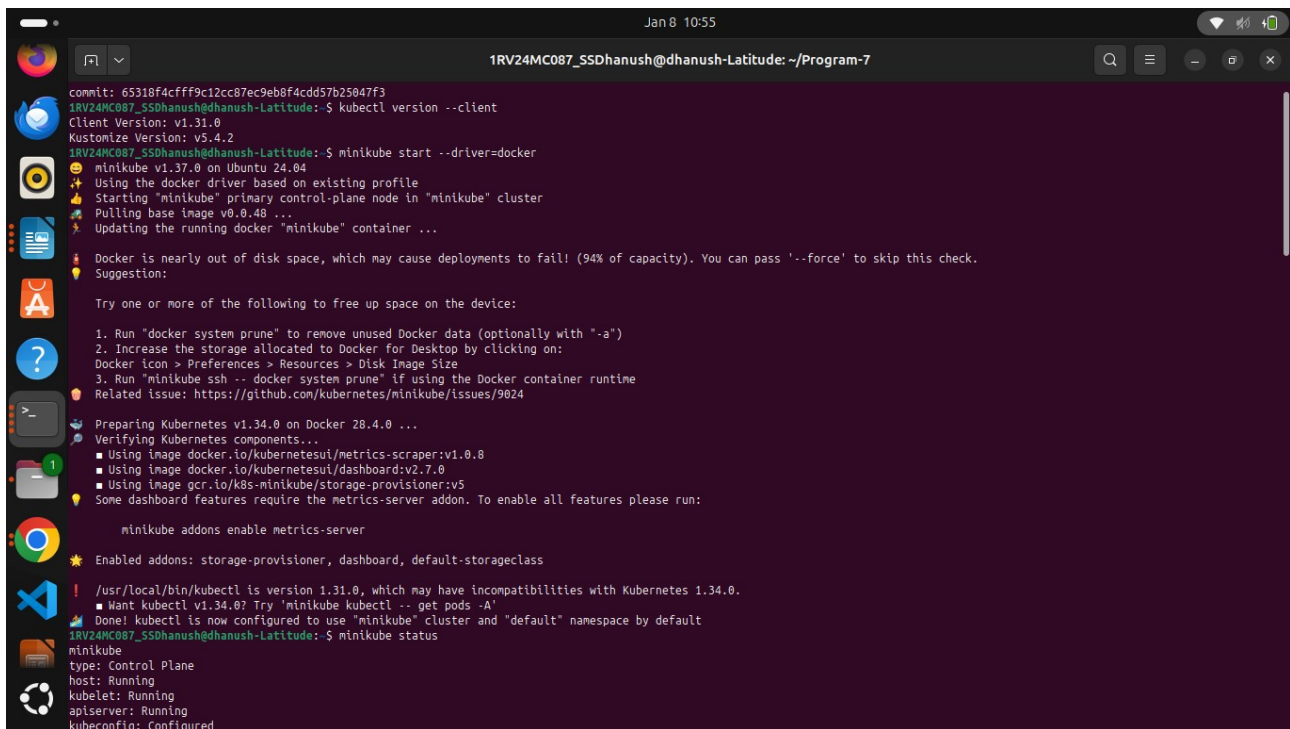
STEP 1: Start Minikube

```
minikube start --driver=docker
```

STEP 2: Check Minikube Status

```
minikube status
```

STEP 3:



```
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25847f3
1RV24MC087_SSDhanush@dhanush-Latitude: $ kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
1RV24MC087_SSDhanush@dhanush-Latitude: $ minikube start --driver=docker
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Updating the running docker "minikube" container ...

Docker is nearly out of disk space, which may cause deployments to fail! (94% of capacity). You can pass '--force' to skip this check.
Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
Related issue: https://github.com/kubernetes/minikube/issues/9024

Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  Using image docker.io/kubernetes/metrics-scraper:v1.0.8
  Using image docker.io/kubernetes/dashboard:v2.7.0
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
Some dashboard features require the metrics-server addon. To enable all features please run:

minikube addons enable metrics-server

Enabled addons: storage-provisioner, dashboard, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC087_SSDhanush@dhanush-Latitude: $ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 4: Create Project Folder

```
mkdir Program-7
```

```
cd Program-7
```

STEP 5: Create Deployment File

```
nano nginx-deployment.yaml
```

```

cd Program-7
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ nano nginx-deployment.yaml
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80

```

STEP 6: Apply the Deployment

```
kubectl apply -f nginx-deployment.yaml
```

TEP 7: Check Deployment & Pods

```
kubectl get deployments
```

```
kubectl get pods
```

```

1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get deployments
kubectl get pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           11s
NAME                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-4lzbh  0/1     ContainerCreating   0           11s
nginx-deployment-6f9664446b-n4pqx  0/1     ContainerCreating   0           11s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           17s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-4lzbh  0/1     ContainerCreating   0           21s
nginx-deployment-6f9664446b-n4pqx  0/1     ContainerCreating   0           21s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    1/2     2            1           34s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-4lzbh  0/1     ContainerCreating   0           39s
nginx-deployment-6f9664446b-n4pqx  1/1     Running             0           39s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           52s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-4lzbh  1/1     Running   0           55s
nginx-deployment-6f9664446b-n4pqx  1/1     Running   0           55s

```

STEP 8: Expose Deployment as a Service

```

kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80

```

STEP 9: Check Service

```
kubectl get services
```

```

nginx-deployment-6f36d444b6-n4pqx 1/1 Running 0 55s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP   10.96.0.1     <none>         443/TCP          2m55s
nginx-deployment NodePort    10.108.1.99   <none>         80:30690/TCP     13s
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30690

```

🌐 Opening service default/nginx-deployment in default browser...
1RV24MC087_SSDhanush@dhanush-Latitude:~/Program-7$ Opening in existing browser session.

```

STEP 10: Access the Web Application

minikube service nginx-deployment

